

Introduction to Macroeconomics (S102016CR)

Spring Semester

<https://moodle.unige.ch/course/view.php?id=6618>

Marcelo Olarreaga, Office M5270

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National Accounts  
and the Real  
Economy"] --> B["Measuring a  
nation's income  
(Chapter 1)"]; A --> C["Measuring the cost  
of living  
(Chapter 2)"]; A --> D["Unemployment  
(Chapter 3)"]
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First Part:
National Accounts
and the Real
Economy

Measuring a
nation's income
(Chapter 1)

Measuring the cost
of living
(Chapter 2)

**Unemployment
(Chapter 3)**

Week 3:

Unemployment

- The objective of this first part of the course has been to measure and understand the determinants of living standards.
- We have so far assumed full employment of all factors of production and ignore unemployment.
- Losing a job is a distressing event in a person's life, not only because of earning losses but also of social status and self-esteem.
- It also implies that the economy is not using all its potential. Workers that could be contributing to the production of goods and services in this economy are not doing so, and we may be hitting diminishing returns much faster than under full employment. Okun's law:

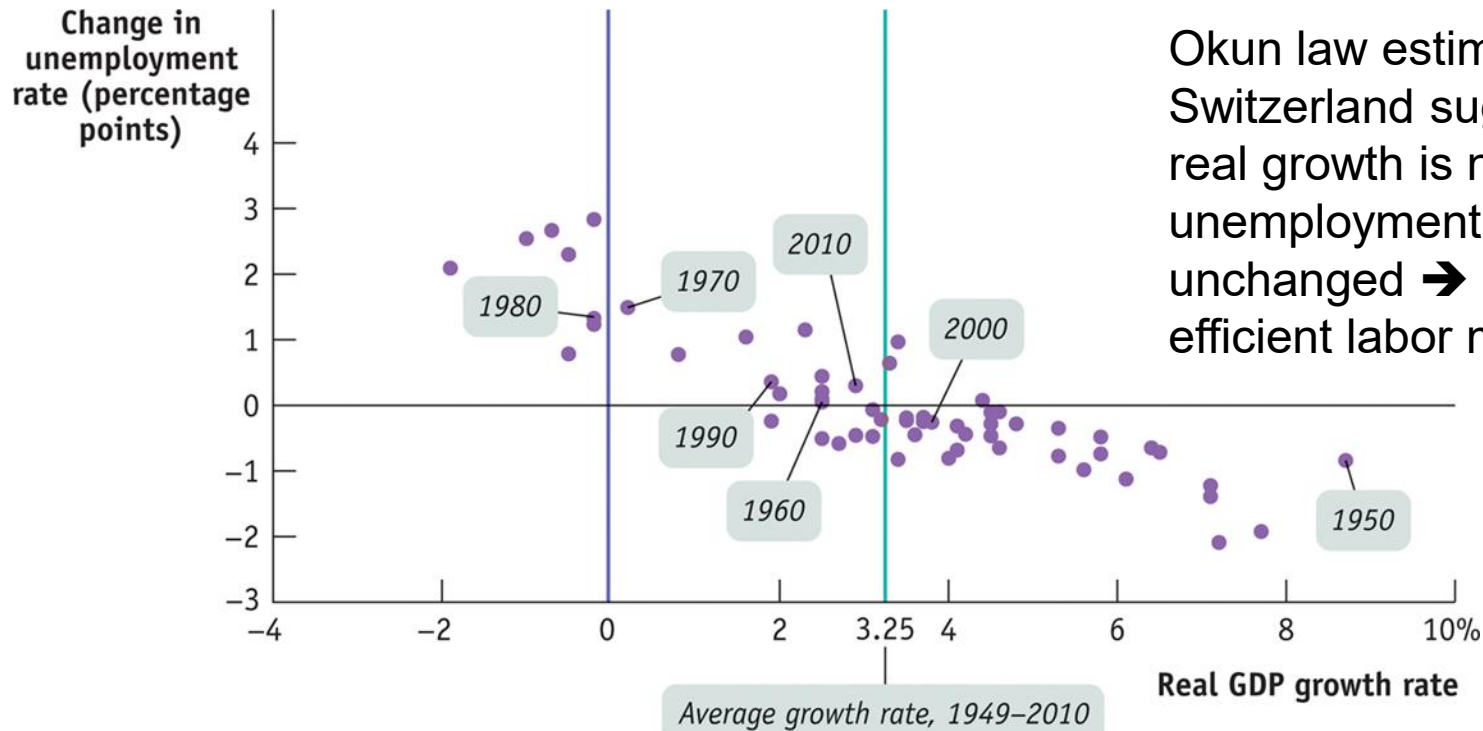
$$\text{Real income growth \%} = 3\% - 2 \times \text{Change in unemployment rate}$$

- Thus, unemployment does not only mean a reduction of living standards at the individual level but also at the macro level.
- This chapter will examine the determinants of unemployment and the role of the government.

Okun's Law in the US

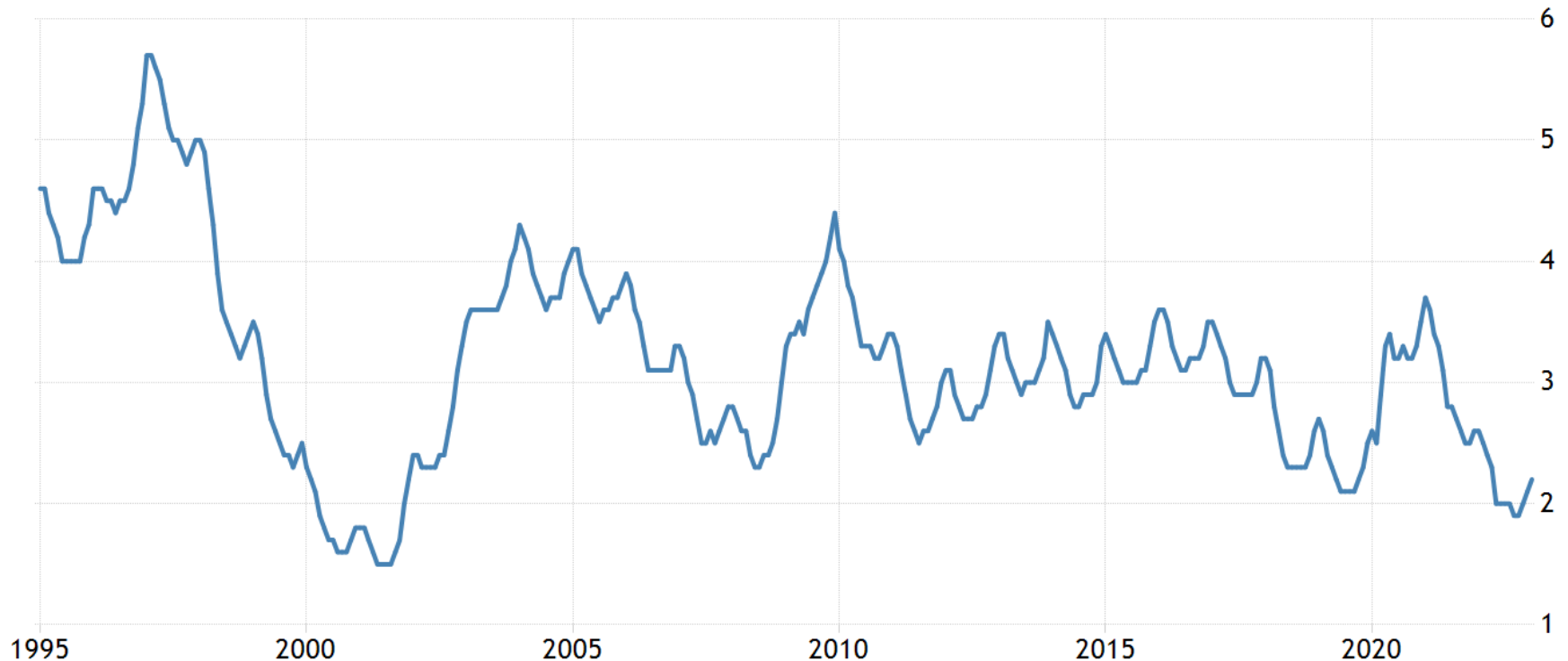
Real growth = 3% - 2 * Change in Unemployment rate

➔ Change in unemployment rate = (3% - Real growth)/2, i.e.
with 3% real growth, unemployment is unchanged.



Source : Krugman & Wells

Is unemployment a problem in Switzerland?



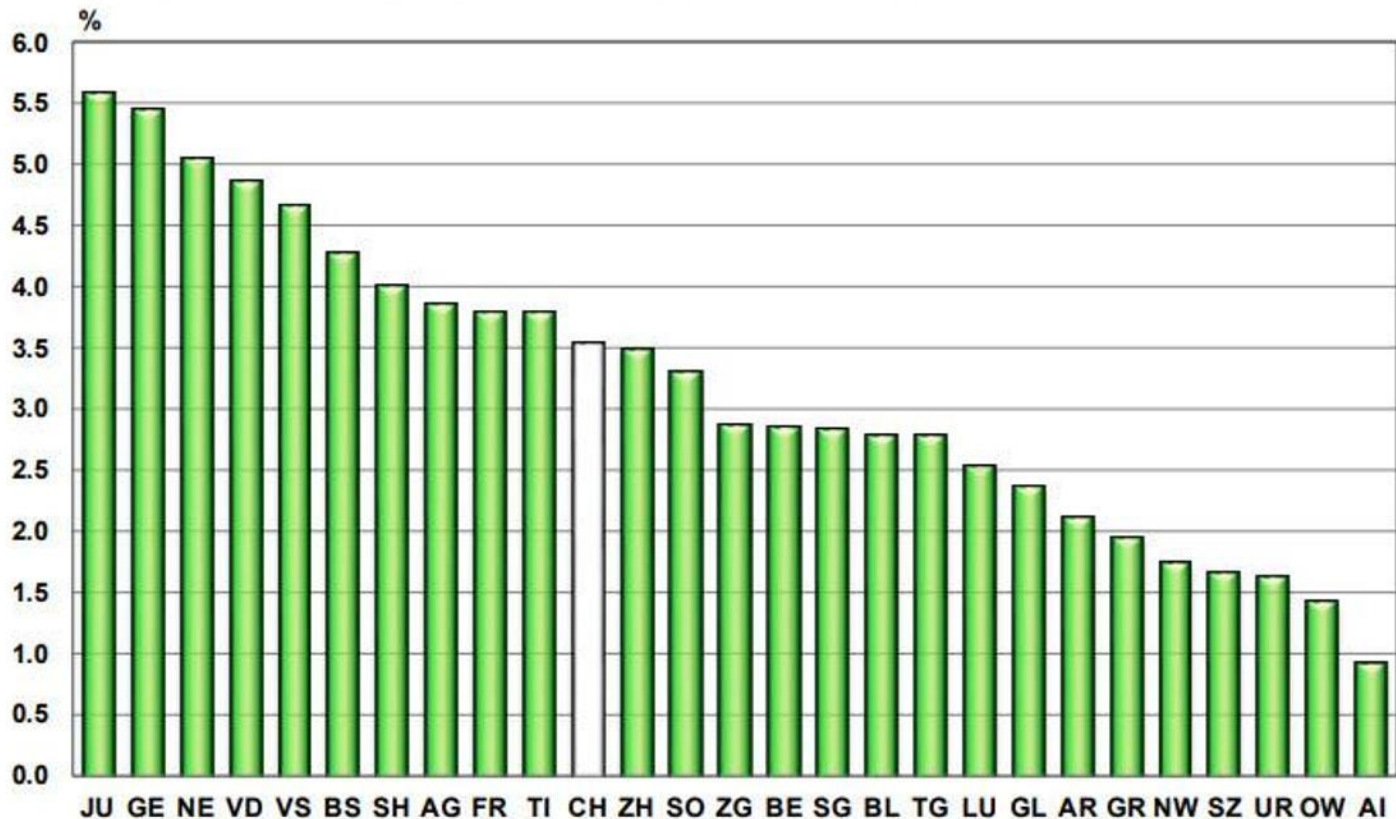
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With an average unemployment rate of 3% over the last decade, Switzerland does relatively well compared to other countries. Average levels of unemployment remained relatively modest during the burst of the dotcom bubble, the 2007-2009 financial crisis or the COVID crisis.

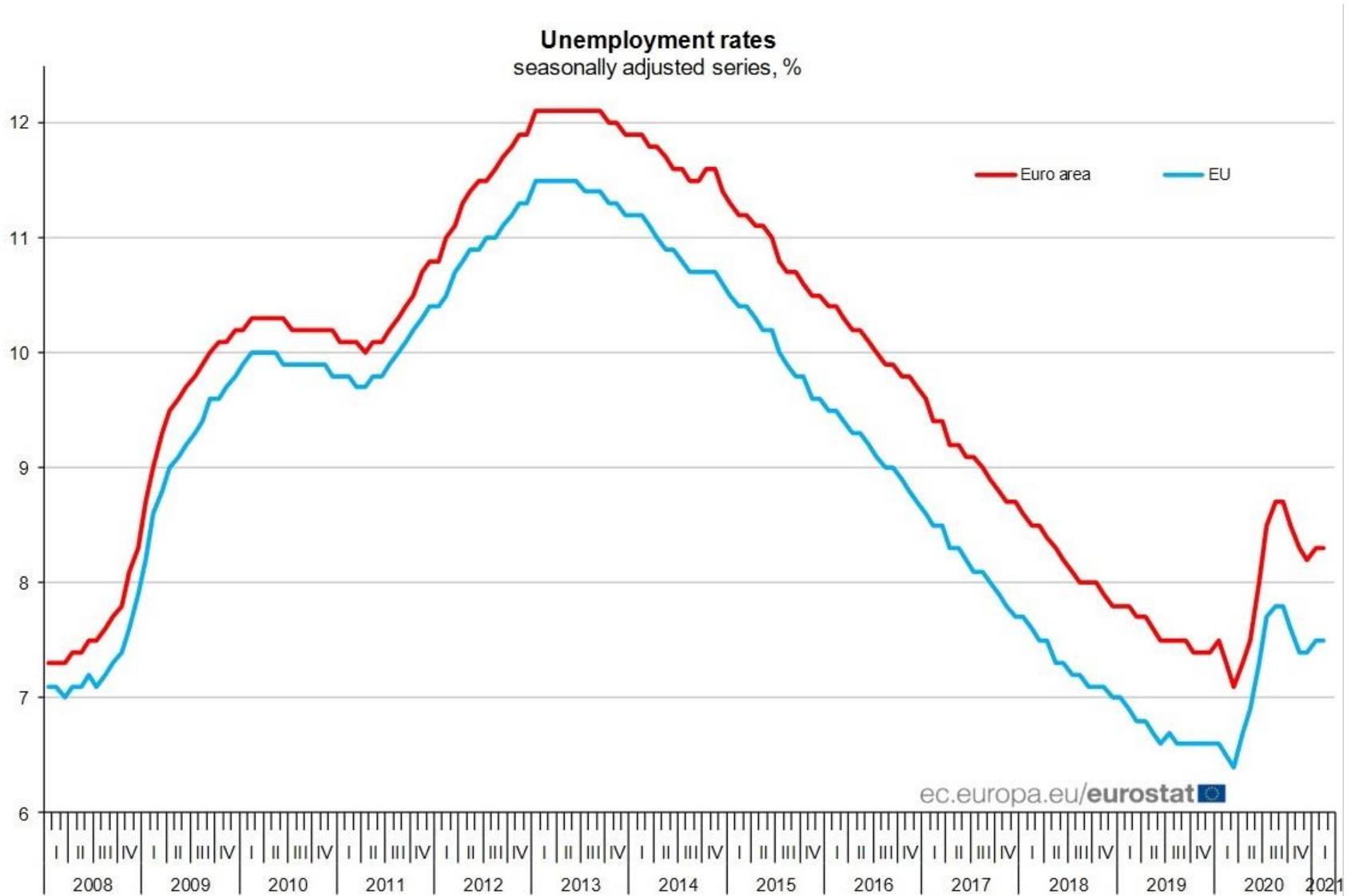
Important differences across cantons in Switzerland

Jura and Geneva were close to 5.5% in December 2020, whereas Appenzell was below 1%

G6: Taux de chômage par canton, décembre 2020



A European perspective



Chapter Contents

- a) Unemployment definition
- b) Measuring Unemployment
- c) Determinants of Unemployment
- d) Government policies
- e) Summary

Chapter Contents

a) Unemployment definition

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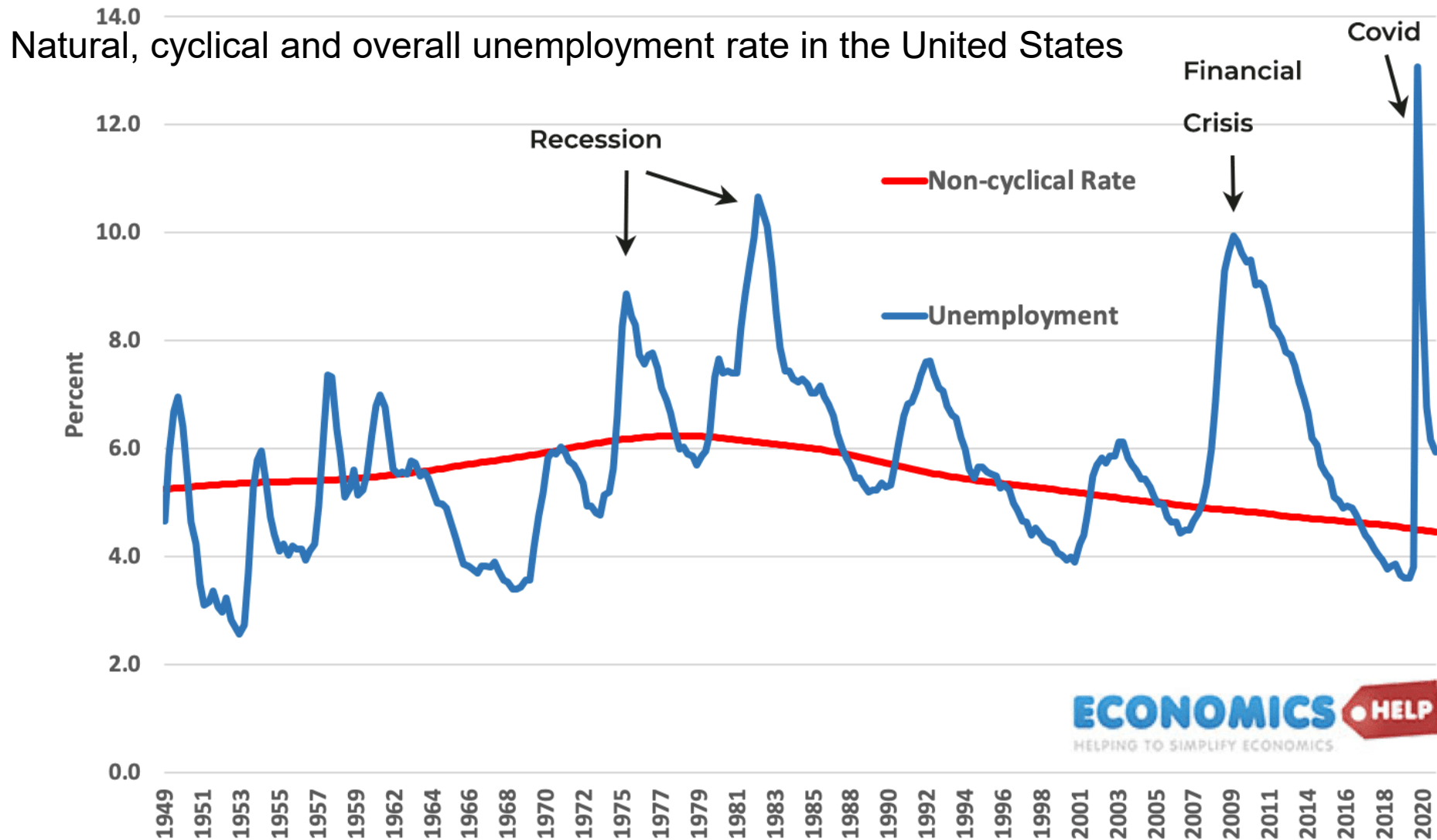
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a. Unemployment definition

Categories of Unemployment

- The problem of unemployment is usually divided into two categories.
- The long-run problem and the short-run problem:
 - The **natural rate** of unemployment, which is the unemployment that does not go away on its own even in the long run.
 - It is the amount of unemployment that the economy typically experiences.
 - The natural rate of unemployment needs to be addressed with long-run policies
 - The **cyclical rate** of unemployment, which is the year-to-year fluctuations in unemployment around its natural rate.
 - It is associated with short-term ups and downs of the business cycle.
 - The unemployment that increases due to cyclical factors has to be addressed with short-run policies

Unemployment and Non-Cyclical Unemployment

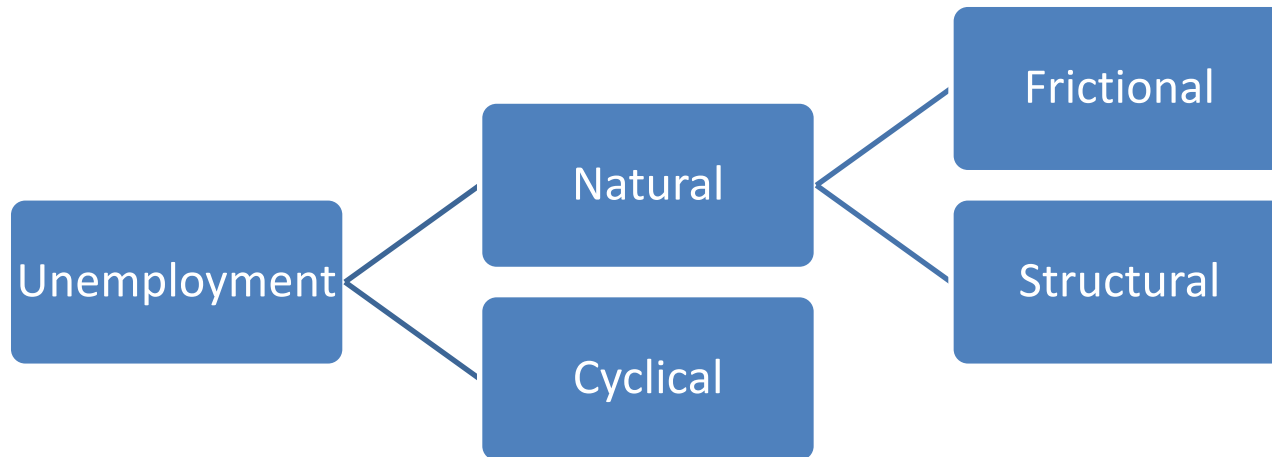


www.economicshelp.org | St Louis Fed - UNRATE and NROU - Accessed 6 Sept 2021

The difference between the overall level of unemployment and the natural level of unemployment (non-cyclical) is cyclical unemployment

- The **natural rate of unemployment** includes **structural** and **frictional** unemployment
 - *Frictional unemployment* refers to the time it takes to match workers with jobs. In other words, it takes time for workers to search for the jobs that best suit their tastes and skills and time for firms to find those workers. This is sometimes referred to as the unemployment caused by **job search** (2010 Nobel Prize for Peter Diamond, Dale Mortensen and Cristopher Pisarrides).
 - *Structural unemployment* is the unemployment that results because the number of jobs available in some labor markets is insufficient to provide a job for everyone who wants one.
- Empirically, it isn't easy to distinguish between the two. Still, one can try to see whether frictional unemployment is getting worse by checking whether the unemployment rate is increasing for a given vacancy rate. This would signal a less efficient functioning of the labor market. We will discuss this using the Beveridge curve during Friday's class

Summing up:



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b. Measuring Unemployment

- Unemployment may be measured in two ways.
 - A monthly survey of households – Labor Force Surveys.
 - The unemployment claimant count.
- Each adult is placed into one of three categories:
 - Employed
 - Unemployed
 - Not in the labor force

The *labor force* is the total number of workers, both employed and unemployed
- The *unemployment rate* is calculated as the percentage of the labor force that is unemployed.

$$\text{Unemployment rate} = U = \frac{\text{Number unemployed}}{\text{Labor force}} \times 100$$

- The **labor force participation rate**:

$$\frac{\text{Labor force}}{\text{Adult population}} \times 100 = \frac{\text{Unemployed} + \text{Employed}}{\text{Adult population}} \times 100$$

- The **employment rate** :

$$\frac{\text{Employment}}{\text{Adult population}} \times 100 = (1 - U/100) * \text{Labor force participation rate}$$

Switzerland 2025: Total population is 8.9 million

**Adult
Population
15-74 years
(6.9 million)**

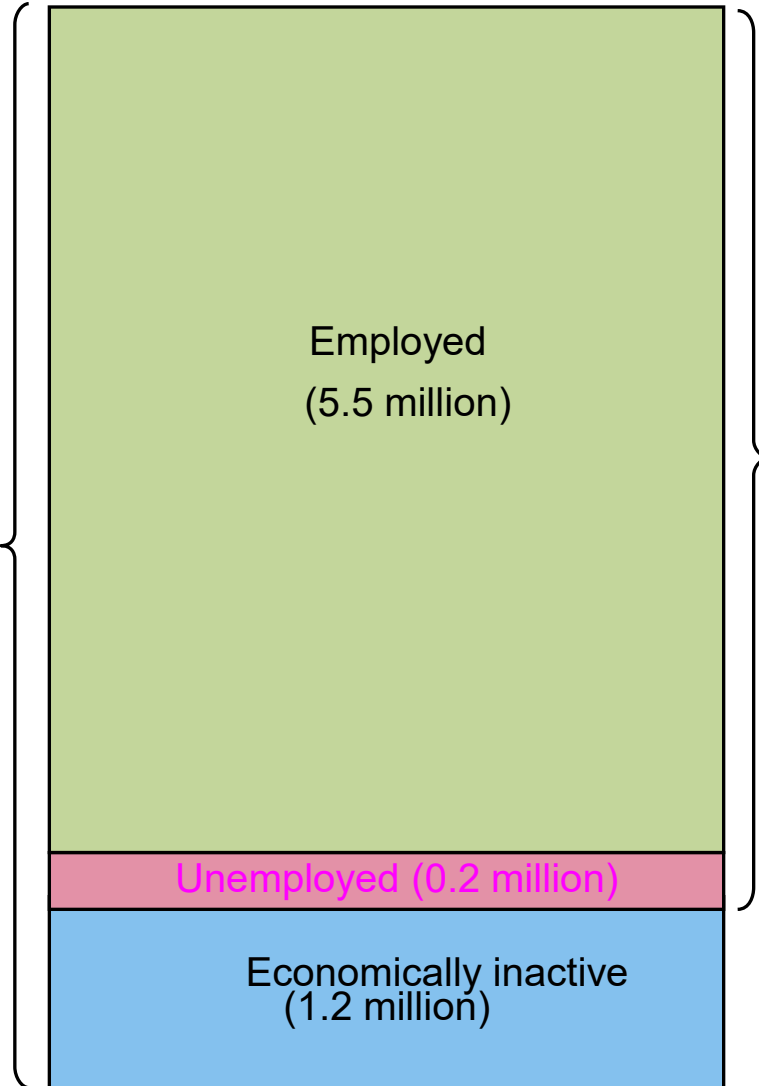
Employed
(5.5 million)

**Labour Force
(5.7 million)**

Unemployed (0.2 million)

Economically inactive
(1.2 million)

Unemployment rate = 3.5%
Labor force participation rate = 83%
(but in full employment equivalent it
is closer to 64%)
Employment rate = 80%



- It is easy to measure the adult population, and those employed, but it is more difficult to distinguish between those unemployed and those inactive.
- Some conventions: The ILO defines an unemployed person as someone between the age of 15 and 74 who:
 - has not been employed (or independently working) over the period under consideration;
 - has been looking for work over the last four weeks;
 - is available for work over the next two weeks
- The population is then surveyed to calculate the unemployment rate.
- In Switzerland, you will be considered unemployed if registered in one of the regional offices (Office Regional de Placement).
 - This includes workers who are working part-time and are looking for a full-time job
 - Because you register it doesn't mean you have the right to receive unemployment benefits.

Does the unemployment rate measure what we want it to?

Not really....

- It is difficult to distinguish between a person who is unemployed and a person who is not in the labor force (economically inactive):
 - *Discouraged workers*, people who would like to work but have given up looking for jobs after an unsuccessful search, don't show up in unemployment statistics.
 - *Free riders*, people may claim to be unemployed to receive financial assistance, even though they aren't looking for work.
- The unemployment rate does not always capture those that are *under-employed* (who have a part-time job and would like a full-time job)

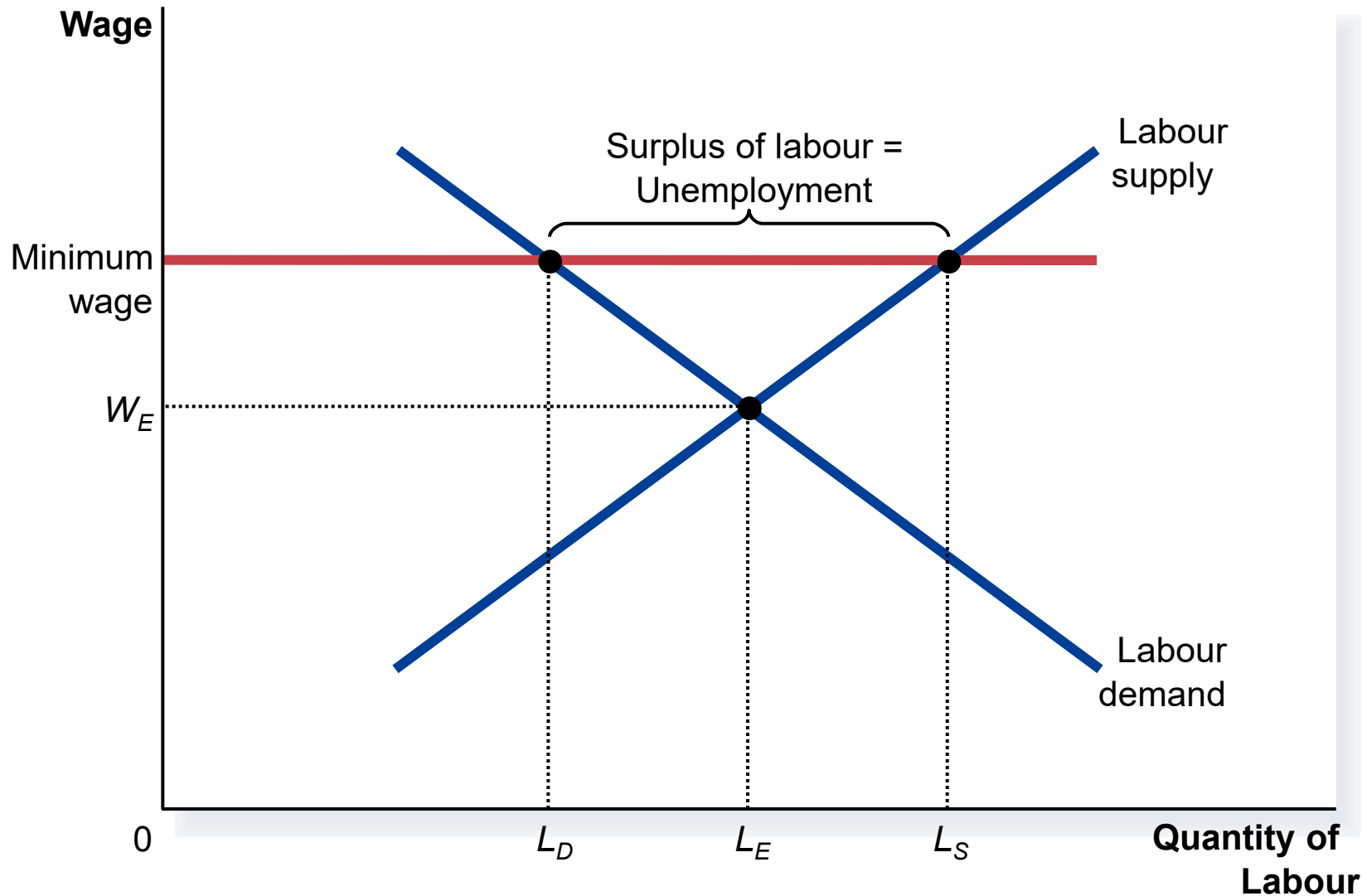
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c. Determinants of unemployment

- **Job search** or frictional unemployment (considered part of the natural unemployment rate) is different from other types of unemployment (structural or cyclical unemployment).
 - It is not caused by a wage rate higher than equilibrium.
 - It is caused by the time spent searching for the “right” job.
- Search unemployment is inevitable because the economy is always changing. Changes in the composition of demand among industries or regions are called sectoral shifts.
- It takes time for workers to search for and find jobs in new sectors.
- **Structural unemployment** is explained by wages that are higher than equilibrium wages due to: minimum wage laws, labor unions, efficiency wages, fair wages

- When the minimum wage is set above the level that balances supply and demand, it creates unemployment.



- A *labor union* is a worker association that bargains with employers over wages and working conditions.
- A union is a type of cartel attempting to exert its market power in the labor market to balance the market power of employers
- The process by which unions and firms agree on the terms of employment is called *collective bargaining*.
- A *strike* will be organized if the union and the firm cannot agree. A strike refers to when the union organizes a withdrawal of labor from the firm.

- **Critics of labor unions** argue that unions cause labor allocation to be inefficient and inequitable.
 - Wages above the competitive level reduce the quantity of labor demanded and cause unemployment.
 - Some workers benefit at the expense of other workers.
- **Advocates of labor unions** contend that unions are a necessary antidote to the market power of firms that hire workers.
 - Higher wages (especially at the bottom of the distribution, i.e., minimum wages) can create, not reduce, employment by curbing firms' market power in labor markets – We will see this in the Friday class
 - They claim that unions are essential for helping firms respond efficiently to workers' concerns.
 - Happier workers are more productive workers
 - Higher minimum wages lead to more spending by households that spend a larger share of their income, boosting the economy.

- *Efficiency wages* are above-equilibrium wages paid by firms to increase worker productivity.
- The theory of efficiency wages states that firms operate more efficiently if wages are above equilibrium.
- A firm may prefer higher than equilibrium wages for the following reasons:
 - Worker Health: Better-paid workers eat a better diet and thus are more productive.
 - Worker Turnover: A higher-paid worker is less likely to look for another job.
 - Worker Effort: Higher wages motivate workers to put forward their best effort.
 - Worker Quality: Higher wages attract a better pool of workers to apply for jobs.
- The reasons behind the 5-dollar day wage introduced by Ford in 1914
- The *fair wage* and other sociological theories stipulate that unless you pay the worker what he considers a “fair wage” he will not put all his effort (part of the efficiency wage model, but with fairness considerations)

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d. Government Policies

- Government programs can affect the time it takes unemployed workers to find new jobs, i.e., **reducing frictional unemployment**.
- These programs include the following:
 - **Government-run employment agencies** which agencies give out information about job vacancies to match workers and jobs more quickly.
 - **Public training programs** aim to ease the transition of workers from declining to growing industries and to help disadvantaged groups escape poverty.
 - **Unemployment insurance** is a government program that partially protects workers' incomes when they become unemployed.
 - Offers workers partial protection against job losses through partial payment of former wages for a limited time to those laid off.
 - If too generous:
 - It may reduce the search efforts of the unemployed.
 - But it may improve the chances of workers being matched with the right jobs.

- To **reduce structural unemployment**, it will have to evaluate the alternatives to each of its determinants and the higher wages resulting from minimum wages, efficiency wages, and labour unions.
- These high wages often respond to other market failures (incomplete contracts, labour market power by firms, etc.), so intervention may be counter-productive if these other problems are not addressed first.
- Policies to **reduce cyclical unemployment** will mainly include policies that increase aggregate demand in the short run to boost economic activity. We will address these policies in detail in the fourth part of this course.

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e. Summary

- The unemployment rate is the percentage of those who want to work but don't have jobs.
- The Office of National Statistics calculates this monthly in most countries (FSO in Switzerland).
- The unemployment rate is an imperfect measure of joblessness.
- One reason for unemployment is the time it takes workers to search for jobs that best suit their tastes and skills.
- A second reason why the economy always has some unemployment is minimum wage laws.
- A third reason for unemployment is the market power of unions.
- The theory of efficiency wages suggests a fourth reason for unemployment. High wages can improve worker health, lower worker turnover, increase worker effort, and raise worker quality.
- A fifth reason is the theory of "fair wages."
- Most unemployment observed at any given time is attributable to a few people who are unemployed for long periods. But most unemployment episodes are short-span (see Friday session).